
Part 3: Market Failures Practice Problems

Econ 502: Advanced Microeconomics

Asymmetric Information

1. Adverse Selection in Insurance with Two Types

A population of consumers each has wealth $w = 100$. With probability p_i , they suffer a loss of $L = 75$. Consumers have utility $u(z) = \ln(z)$ and so are risk-averse. There are two risk types:

- **Low-risk** consumers have $p_L = 0.2$
- **High-risk** consumers have $p_H = 0.6$

A fraction λ of the population is high-risk. Each consumer knows their own type; the insurer cannot tell types apart. Insurance is “full insurance” at premium P (the insurer pays L if the loss occurs).

- a) For each type $i \in \{L, H\}$, compute the maximum premium $P_i^{\max}$ that makes the consumer just indifferent between buying full insurance and not. (This is the type’s willingness to pay.) Verify that $P_H^{\max} > P_L^{\max}$.

- b) Suppose the insurer charges the **actuarially fair pooled premium**:

$$P_{\text{pool}} = [\lambda p_H + (1 - \lambda)p_L] \cdot L$$

For $\lambda = 0.5$, compute P_{pool} . Which types are willing to buy at this premium? Will the insurer break even?

- c) For what range of λ does the market unravel completely (only high-risk types buy at any premium that an insurer would offer)? Explain the economic logic in words.
- d) Suppose the government mandates that everyone purchase insurance at the pooled premium P_{pool} . Compare each type’s expected utility under the mandate with their expected utility without insurance. Who is better off? Who is worse off? Why might this still be a desirable policy from a social welfare perspective?

2. Signaling Through Education (Spence Model)

A labor market has two types of workers:

- **High-ability (H)**: productivity $w_H = 100$, fraction λ of the workforce
- **Low-ability (L)**: productivity $w_L = 40$, fraction $1 - \lambda$ of the workforce

Workers can choose to obtain $e \geq 0$ years of education. Education is purely a signal — it does not raise productivity. The cost of acquiring e years of education is:

$$c_H(e) = 5e \quad (\text{high types})$$

$$c_L(e) = 15e \quad (\text{low types})$$

Firms observe a worker's education e but not their type, and pay a wage equal to expected productivity given e .

- a) Why is the cost of education lower for high-ability workers? What real-world feature of education does this represent?
- b) **Pooling equilibrium.** Suppose all workers choose $e = 0$ and firms pay the average wage $\bar{w} = \lambda w_H + (1 - \lambda)w_L$. Is this an equilibrium? Under what condition on λ would each type prefer this to revealing their type? (Take $\lambda = 0.5$ for a numerical answer.)
- c) **Separating equilibrium.** Find a level of education e^* such that high types choose e^* and low types choose $e = 0$, with firms paying w_H to anyone with $e \geq e^*$ and w_L to anyone with $e < e^*$. Specifically, derive the range $[e_{\min}, e_{\max}]$ such that:
 - Low types prefer $e = 0$ (and wage w_L) to mimicking e^* (and getting wage w_H)
 - High types prefer e^* (and wage w_H) to $e = 0$ (and getting w_L)
- d) Compute total social welfare under the separating equilibrium of part (c) (using $e^* = e_{\min}$) versus under symmetric information (each type works without education). Is signaling socially wasteful? Explain.

3. Moral Hazard: Principal-Agent Problem

A risk-neutral firm hires a worker who can choose effort $e \in \{0, 1\}$ (low or high). Output y is stochastic and depends on effort:

- If $e = 0$: $y = 100$ with probability 0.4, $y = 0$ with probability 0.6
- If $e = 1$: $y = 100$ with probability 0.8, $y = 0$ with probability 0.2

The worker's utility is $u(w) - c(e)$ where $u(w) = \sqrt{w}$, $c(0) = 0$, $c(1) = 2$. The worker's outside option gives utility $\bar{u} = 5$.

- a) **First-best (effort observable).** Suppose effort is contractible. The firm offers a fixed wage conditional on effort. What is the wage w_1 that induces $e = 1$ at minimum cost? What is the firm's expected profit when it implements $e = 1$ vs. $e = 0$? Which effort level does it want to implement?
- b) **Second-best (effort hidden).** Now effort is unobservable. The firm offers a contract (w_H, w_L) : wage w_H if output is high, w_L if output is low. Write down (i) the **participation constraint** (worker accepts) and (ii) the **incentive compatibility constraint** (worker chooses $e = 1$ over $e = 0$).
- c) Solve for the optimal (w_H, w_L) that implements $e = 1$ at minimum expected wage cost. (Hint: both constraints bind at the optimum.) Verify that $w_H > w_L$ and explain why.
- d) Compare the firm's expected wage bill under first-best and second-best. The difference is the "cost of moral hazard." Explain intuitively why hidden effort forces the firm to expose the worker to risk.

Externalities

4. Pigouvian Tax with Heterogeneous Firms

Two firms produce a chemical, with pollution as a byproduct. Firm i 's private marginal cost of producing q_i units is:

$$MC_1(q_1) = 2q_1, \quad MC_2(q_2) = 4q_2$$

The market price is $P = 40$ per unit. Each unit of output produces 1 unit of pollution, and each unit of pollution causes \$10 of damage to a downstream community.

- Find each firm's privately optimal output q_i^{priv} (where $P = MC_i$). Compute total output and total pollution damage.
- Find the socially efficient output q_i^* for each firm (where $P = MC_i + 10$). Compute total output, total damage, and the deadweight loss from unregulated production.
- Pigouvian tax.** The regulator imposes a per-unit tax $t = 10$ on output of both firms. Show that the tax decentralizes the efficient outcome.
- Uniform quota.** Suppose instead the regulator imposes a single quota $\bar{q}$ on each firm, where total output equals the efficient total from part (b). Compute $\bar{q}$. Does the uniform quota replicate the efficient allocation? If not, compute the welfare loss relative to the Pigouvian tax. Explain why a uniform regulation is generally inferior when firms differ.

5. Cap-and-Trade with Heterogeneous Abatement Costs

Two power plants currently emit a total of 200 tons of CO_2 . The marginal cost of abatement (reducing emissions by one ton) is:

$$MAC_1(a_1) = 2a_1, \quad MAC_2(a_2) = a_2$$

where a_i is the number of tons abated by plant i . Society's marginal damage from each ton of emissions is constant at \$80.

- Efficient abatement.** Find the socially efficient total abatement and the efficient allocation (a_1^*, a_2^*) across the two plants. (Hint: at the efficient allocation, $\$MAC_1 = MAC_2 = \$$ marginal damage.)
- Uniform standard.** Suppose the regulator requires each plant to abate the same amount such that total abatement equals the efficient total from part (a). Compute the total abatement cost. How does it compare to the efficient total abatement cost?
- Cap-and-trade.** The regulator issues a number of permits equal to total efficient emissions and distributes 100 permits to each plant. Plants can trade permits at price τ . Each plant chooses abatement to minimize $MAC_i(a_i) \cdot a_i$ plus the cost (or minus the revenue) of net permit purchases. Find the equilibrium permit price τ^* and the equilibrium abatement allocation. Verify that the market replicates the efficient allocation.
- Initial allocation neutrality.** Suppose instead the regulator gives all permits to Plant 1 (the high-cost abater). Show that the equilibrium abatement allocation is unchanged. What does change? Connect this result to the Coase theorem.

6. Coase Theorem with Reciprocal Externalities

A bakery and a doctor share a building. The bakery's machinery generates noise. Profits depend on whether the bakery uses noisy or quiet machinery and whether the doctor's office is soundproofed:

	Doctor not soundproofed	Doctor soundproofed
Bakery noisy	$\pi_B = 200, \pi_D = 60$	$\pi_B = 200, \pi_D = 100$
Bakery quiet	$\pi_B = 150, \pi_D = 130$	$\pi_B = 150, \pi_D = 130$

Soundproofing costs the doctor \$30 (reflected above). Switching to quiet machinery costs the bakery \$50 in foregone profit.

- Compute total surplus $\pi_B + \pi_D$ for each cell. Which combination is socially efficient?
- Right to silence.** Suppose the doctor has the legal right to a quiet office. Without bargaining, what does the bakery do? With Coasian bargaining, what is the efficient outcome and what is the range of feasible payments from doctor to bakery?
- Right to be noisy.** Suppose the bakery has the legal right to make noise. Without bargaining, what does the doctor do? With Coasian bargaining, what is the efficient outcome and what is the range of feasible payments?
- Compare your answers in (b) and (c). What does this illustrate about the Coase theorem? Why might the assignment of property rights still matter in practice?

Public Goods

7. Public Good Provision with Three Contributors

Three neighbors share a small park. Each has income $Y_i = 60$ and chooses how much to contribute $g_i \geq 0$ to park maintenance. Total park quality is $G = g_1 + g_2 + g_3$. Each neighbor's utility is:

$$U_i(x_i, G) = x_i \cdot G$$

where $x_i = Y_i - g_i$ is private consumption.

- Derive each neighbor's best response function $g_i^*(g_{-i})$, where $g_{-i} = \sum_{j \neq i} g_j$. By symmetry, find the Nash equilibrium contributions g_i^* and total provision G^* .
- Social optimum.** A planner chooses (x_1, x_2, x_3, G) to maximize $\sum_i U_i$ subject to $\sum_i x_i + G = \sum_i Y_i$. Find G^{**} and the equal-private-consumption allocation x_i^{**} . (Hint: with utility $x_i G$, total welfare is XG where $X = \sum x_i$.)
- Verify that the Nash equilibrium satisfies $MRS_i = 1$ for each i but the social optimum satisfies $\sum_i MRS_i = 1$ (the **Samuelson condition**). Compute the ratio G^*/G^{**} . As the number of contributors N grows, what happens to the fraction of the efficient public good that is voluntarily provided?

- d) Suppose each contributor receives a subsidy s on their contribution: a contribution of g_i costs them only $(1 - s)g_i$, with the government covering the rest. Find the subsidy rate s^* that decentralizes the efficient public good level G^{**} .

8. Lindahl Prices

Two consumers fund a public good G . They have utility:

$$U_1(x_1, G) = x_1 + 10 \ln G, \quad U_2(x_2, G) = x_2 + 6 \ln G$$

Each unit of G costs \$1 to produce (so the resource constraint is $x_1 + x_2 + G = Y_1 + Y_2$). Both have ample income.

- a) **Samuelson condition.** Compute each consumer's $MRS_i = (\partial U_i / \partial G) / (\partial U_i / \partial x_i)$ and find the efficient G^{**} .
- b) **Lindahl prices.** Suppose each consumer is charged a personalized price-per-unit τ_i for the public good, where $\tau_1 + \tau_2 = 1$ (the prices add up to the marginal cost of G). Each consumer takes τ_i as given and chooses how much G they want to “demand,” then the level of G is set so both demand the same quantity. Find the Lindahl prices (τ_1^*, τ_2^*) that support the efficient G^{**} .
- c) Verify that at the Lindahl prices, each consumer “demands” exactly G^{**} . Why does this mechanism elicit efficient provision in theory?
- d) Why is implementing Lindahl pricing difficult in practice? What incentive do consumers have when reporting their preferences to the planner?